IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25-26J-175

1. Topic (12 words max)

An Integrated Smart System for Cashew Cultivation

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Cashew cultivation faces challenges such as soil suitability assessment pest and disease management, and post harvesting grading. Traditional methods are often time consuming and prone to human errors. This research project aims to develop an integrated smart system that leverages image processing and data analysis to address these issues [5].

1. Soil suitability assessment:
Cashew trees thrive in well drained, sandy loam soils with a pH of 6.0-6.5. However, farmers often lack tools to access soil conditions accurately [1]. Traditional soil testing is time consuming and requires laboratory facilities, which may be inaccessible in rural areas. Poor soil conditions may lead to poor growth, low yield, and increased sensitivity towards to diseases.

2. Manual cashew grading inefficiencies:
After harvesting the grading process is labor subjective, relying on visual inspection by workers. Inconsistent grading leads to price fluctuations in market, affecting farmers' income. There is no standardized, automated system to evaluate cashew nuts based on weight, size, color, and density [2].

3. Disease detection and management:
Common cashew diseases such as Anthracnose, Powdery Mildew cause significant yield losses. Farmers often detect diseases too late due to a lack of early diagnostic tools [3]. Misdiagnosis leads to improper pesticide use, increasing costs and environmental harm.

4. Pest detections and management:

Pests like Tea mosquito Bugs and Stem borers damage leaves, nuts, and stems. Farmers rely on chemical sprays, which are often overused due to lack of precise pest identification [4]. Overuse of pesticides harms beneficial insects and increases production costs and lack of knowledge in pesticides may leads to unwanted increased costs.

References

[1] R. R. Nair, K. P. Chandran, and S. Thomas, "Soil suitability evaluation for cashew (Anacardium occidentale L.) in lateritic soils of Kerala," Journal of Plantation Crops, vol. 45, no. 2, pp. 102–110, 2017.

[2] FAO, "Global standards for cashew grading and quality control," Food and Agriculture Organization, Rome, Tech. Rep. 12, 2020.

[3] M. L. Deadman et al., "Anthracnose and powdery mildew: Emerging threats to cashew production in East Africa," Plant Pathology Journal, vol. 38, no. 3, pp. 221–230, 2019.

[4] P. S. Bhat and K. V. Saji, "Integrated pest management in cashew: Challenges and solutions," Journal of Applied Entomology, vol. 144, no. 8, pp. 678–689, 2020.

[5] A. Kamilaris et al., "Deep learning in agriculture: A survey," Computers and Electronics in Agriculture, vol. 147, pp. 70–90, 2018.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Existing solutions address cashew cultivation challenges in isolation like lab-based soil tests or manual pest scouting. However, an integrated AI-driven system leveraging image processing, machine learning, and data analytics can provide a cost-effective and scalable solution:

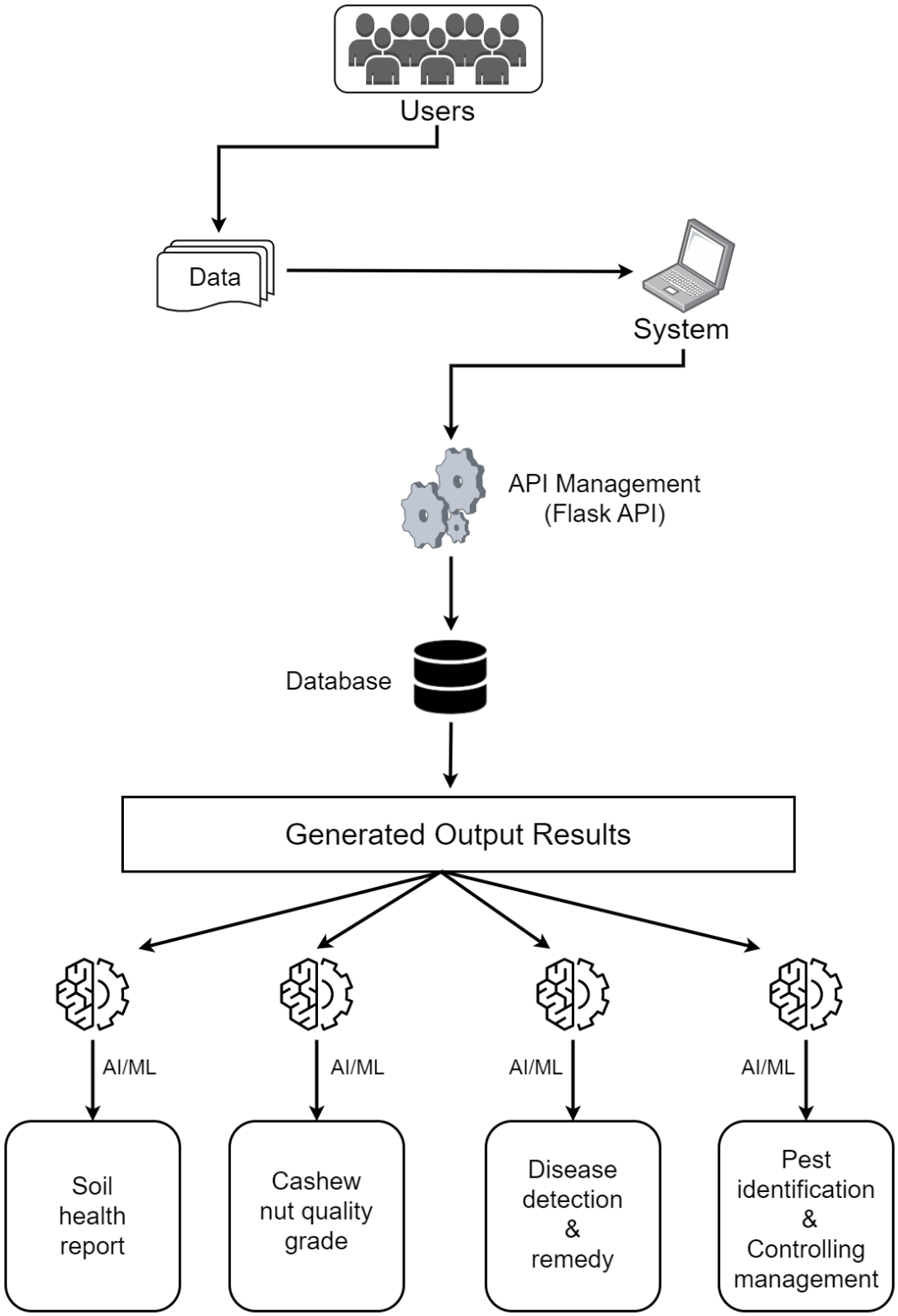
1.Soil analysis:
A deep learning model will classify soil type, moisture, and health indicators (color, texture) using images captured via smartphones or low-cost cameras. It eliminates the need for expensive lab tests by providing instant, on-field soil health reports.

2. Automated cashew grading:
A computer vision-based algorithm will assess cashew nut quality by analyzing size, color, and defects from images. estimation can be derived from size density correlations in the dataset, avoiding the need for physical sensors.

3. Disease detection and remedy:
A deep learning model (CNN) will be trained to classify common cashew diseases such as Anthracnose, Powdery Mildew, Leaf Weight minor damage. The deep learning model architecture will process high resolution leaf and nuts images to detect disease symptoms. The model will be trained on a curated dataset of diseased and healthy leaves with manually annotated data.

4. Pest detection and controlling:
Object detection model will identify tea mosquito bug, stem and root borers, leaf-eating caterpillars, then the system will count pest instances per leaf area, estimate infestation severity. An integrated system will recommend treatments which includes organic treatments, chemical controls, cultural practices.

Conceptual Diagram



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain expertise requirement
Understanding of cashew cultivation practices, soil requirements, physical status of cashew, common diseases and pests.
Proficiency in CNN architectures (ResNet, EfficientNet) for image classification, experience with object detection models (YOLO, Faster R-CNN) for pest identification and need skills in image preprocessing (OpenCV) and model optimization (TensorFlow/PyTorch).
Using agronomic parameters to analyze the soil health trends and correlate image features, statistical modeling for cashew grading.

Knowledge requirement
Team should understand how the Convolutional Neural Networks (CNN) and object detection models (YOLO, Faster R-CNN) can be adapted for agricultural use cases. This includes understanding in soil variations (laterite/ red soil), seasonal disease patterns, pest lifecycles, local farming practices, and post-harvest handling standards. Also ensuring the system's recommendations align with local conditions and available resources, making the AI solutions practical for farmers across diverse growing regions.

Data requirements

- 5000+ labeled images of different soil types (sandy, loamy, clay) and conditions (dry, moist, eroded). pH levels of the soils in different geographic location.
- 3000+ images of graded nuts with annotations for size, color, and defects. For weight and density measurements through trading regression models.
- 4000+ annotated leaf images covering all the diseases and pests related to cashew cultivation.
- List of appropriate recommendations of treatments linked with the respective disease and pest types.

8. Objectives and Novelty

Main Objective Develop an integrated AI system to optimize cashew cultivation through image-based soil analysis, automated nut grading, disease detection with remedy, and pest identification and pest controlling management. The system will reduce reliance on manual labor and improving yield quality.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Mithushh P IT22180902	Development of soil health analysis system for cashew plants.	• Learn what each cashew cultivar needs (water, soil, etc.). • Check soil properties like color and texture. • Use machine learning to match soil with the best cashew type. • Recommend the right cashew type for each soil to help farmers.	• Using Machine Learning to Match the Best Cashew Type with the Right Soil.
Sanoch M IT22159694	Development of automated grading system for cashew nuts.	• Create dataset with cashew nut images with labels. • Train a model to detect the defects. • Extract the nut dimensions from the model and predict the density using measurement ratios and shape. • Validate the measurements by comparing with manual scale measurements. • Build user-interface for farmers to upload images and assign grades for nuts.	• Grade the cashew nuts based on size, color, texture defects, and density with the use of industry standards.

Thimothy K L IT22206428	Integrated disease detection system for cashew plants.	• Create a dataset with images of diseased leaves with labels. • Build a suitable model to detect the disease. • Fine tune the model for high accuracy rate. • Build user-interface for farmers to upload images and get the details of the diseases.	• Fast and accurate early disease identification and classification to prevent spread to the entire plant.
Usman M.H IT22351104	Integrated pest identification and controlling system for cashew plants.	• Capture real time images of pests at different life cycles and create a dataset with labels. • Build a model to detect the pests. • Fine tune the model for high accuracy rate. • Suggest appropriate pest controlling methode according to the pest identification results. • Build user-interface for farmers to upload images and receive suggestion on pest controlling.	• A fast AI-powered system that detects pests on cashew plants, recommends suitable organic or inorganic pesticides, and protecting beneficial insects.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Mithush P IT22180902	The soil health analysis component applies AI and deep learning to classify soil using a hybrid CNN. It integrates computer vision, model optimization, and real-time prediction via a user interface. This aligns with the IT specialization by combining machine learning, image processing, and full-stack development to support precision agriculture.
Sanoch M IT22159694	This component applies IT specialization through machine learning, computer vision, and full-stack development. It uses deep learning to detect nut defects, predict weight from dimensions, and automate grading. A custom user interface integrates AI with real-time use, demonstrating skills in model training, image analysis, and intelligent system design.
Thimothy K L IT22206428	This component reflects IT specialization through AI-driven plant disease detection using image classification models. It combines deep learning, data handling, and intelligent decision systems. Integrated with the user interface, it delivers real-time, accurate solutions—demonstrating skills in model tuning, system integration, and practical application of machine learning.
Usman M.H IT22351104	This component applies IT specialization by using machine learning and computer vision to identify pests across life stages. It integrates an AI model with a user interface to automate pest detection and suggest control methods. It demonstrates expertise in data science, model tuning, intelligent decision-making, and real-time user-centered software development.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Shashika	Lokuliyana	
Co-Supervisor	Ms.	Suranjini	Silva	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended for project proposal.

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Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

1) Reword the objectives and accordingly rewrite the one novelty component.

Staff Member's Name	Signature
Shashika Lokuliyana	
Suranjini Silva	

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.